did in the 1970s, and the first Bank Swallows reach their destination 25 days sooner than previously. However, there is evidence that birds migrating from Central America to New England in the US have declined faster than birds that remain in New England all year. This is probably because the migrant birds have been unable to adjust their departure dates from Central America to arrive in time to benefit from the earlier abundance of insects the way local birds do.

Climate change also appears to have changed the behavior of hibernating mammals. Zoologists at the Rocky Mountain Biological Laboratory found that yellow-bellied marmots living in Colorado emerged 38 days earlier in 1999 than they had done in 1975. In 2012, scientists at the University of Alberta found that in the last two decades, late snowfall has delayed the emergence of the Rocky Mountain ground squirrel from hibernation by 10 days. This has cut down the already short active period in which they mate, give birth, and feed to prepare for the next hibernation cycle.

Decoupling

Some organisms' survival could be threatened by the "decoupling" of interactions between species. This could seriously upset the balance of ecosystems. If flowers appear earlier, the bees that pollinate them can respond in one of two ways: they can emerge earlier; or they can move to a higher latitude to match later flower emergence farther from the equator. Studies of 10 wild bee species in northeast North America have shown that their behavior has changed in line with earlier flowering. However, bumblebees in Colorado have not matched the changes and their population has fallen. If pollinators decline, so may the plants that they pollinate.

There is evidence that many primary consumers have adjusted to changed natural phenomena, but species higher in the food chain seem to find it harder to make the change. Although birds are now nesting earlier than they once did, the timing of insect emergence has advanced more